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Method of manufacturing spliced display screen and spliced display screen

Abstract

The present application provides a method of manufacturing a spliced display screen and a spliced display screen. The method of manufacturing a spliced display screen includes: providing a glass substrate and a display panel, wherein the display panel includes a flexible substrate and a display unit, and the display panel includes a bonding area and a non-bonding area; attaching a first protective film to a side of the display unit away from the glass substrate; disposing a supporting glass on a side of the first protective film away from the flexible substrate; separating the glass substrate from the flexible substrate; cutting the first protective film, the display panel, and the supporting glass located in the non-bonding area to obtain a plurality of first components; providing a second protective film, and splicing the plurality of first components on a surface of a side of the second protective film.

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Background/Summary

BACKGROUND OF INVENTION

Field of Invention

(1) The present application relates to the field of display technology, and in particular to a method of manufacturing a spliced display screen and a display screen.

Description of Prior Art

(2) Light-emitting diodes (LEDs) are semiconductor electronic components that can convert electrical energy into light energy. Because of their small sizes, long service life, colorful colors, and low energy consumption, the light-emitting diodes widely are used in fields of lighting, display screens, signal lights, backlight sources, toys, and so on. A mini-LED is also known as sub-millimeter light-emitting diode, usually having a size of 80 micrometers to 200 micrometers, which is a new generation of LED technology and inherits characteristics of small-pitch LEDs, such as high efficiency, high reliability, high brightness, fast response times, and lower power consumption and cost.

(3) In recent years, flexible electronic products have attracted widespread attention around the world and have been rapidly developed. When making large-sized screens in the flexible electronic products, a plurality of flexible devices need to be spliced together. However, the existing flexible devices have a problem of great difficulty in splicing.

SUMMARY OF INVENTION

(4) Embodiments of the present application provide a method of manufacturing a display module and a display panel, which solves the problem of splicing difficulty in existing flexible devices.

(5) An embodiment of the present application provides a method of manufacturing a spliced display screen, and the manufacturing method includes:

(6) providing a glass substrate and a display panel, wherein the display panel includes a flexible substrate and a display unit sequentially formed on a surface of a side of the glass substrate, and the display panel includes a bonding area and a non-bonding area;

(7) attaching a first protective film to a side of the display unit away from the glass substrate;

(8) disposing a supporting glass on a side of the first protective film away from the flexible substrate;

(9) separating the glass substrate from the flexible substrate;

(10) cutting the first protective film, the display panel, and the supporting glass located in the non-bonding area to obtain a plurality of first components;

(11) providing a second protective film, and splicing the plurality of first components on a surface of a side of the second protective film, wherein the second protective film is disposed on a side of the flexible substrate away from the first protective film.

- (12) Optionally, the step of splicing the plurality of first components on the surface of the side of the second protective film includes:
- (13) disposing an end of one of the first components away from the bonding area adjacent to an end of another one of the first components away from the bonding area.
- (14) Optionally, after the step of splicing the plurality of first components on the surface of the side of the second protective film, the manufacturing method further includes:
- (15) separating the supporting glass from the first protective film.
- (16) Optionally, the manufacturing method further includes:
- (17) electrically connecting a chip-on-film to one end of each of the flexible substrate located in the bonding area; and
- (18) electrically connecting a printed circuit board to an end of each of the chip-on-film away from the flexible substrate.
- (19) Another embodiment of the present application also provides a spliced display screen, wherein the spliced display screen includes:
- (20) a second protective film;
- (21) a plurality of first components arranged adjacently, wherein the plurality of first components are arranged on a same side of the second protective film to form a plurality of second components, each of the first components includes a display panel and a first protective film that are stacked, the display panel is disposed between the first protective film and the second protective film, wherein the display panel includes a flexible substrate and a display unit that are stacked, and the display panel includes a bonding area and a non-bonding area.
- (22) Optionally, an end of the first protective film away from the bonding area is flush with an end of the flexible substrate away from the bonding area.
- (23) Optionally, a number of rows of the plurality of second components spliced on the second protective film is n ; a number of columns of the plurality of second components spliced on the second protective film is 1; and the $n \geq 1$.
- (24) Optionally, the first protective film and the second protective film are both polyethylene terephthalate (PET) films.
- (25) Optionally, the display unit includes one of LED, mini-LED, micro-LED, or OLED.
- (26) Optionally, a chip-on-film is disposed at one end of each of the flexible substrate away from the non-bonding area; and
- (27) a printed circuit board is disposed at one end of each of the chip-on-film away from the flexible substrate.
- (28) Beneficial effects of the present application are: in the method of manufacturing the spliced display screen provided in the embodiments of the present application, the first protective film is attached to the side of the display panel away from the glass substrate; the supporting glass is disposed on the side of the first protective film away from the flexible substrate; then the glass substrate is separated from the flexible substrate; the first protective film and the display panel located in the non-bonding area are cut, and the cut first protective film and the flexible substrate are defined as the first component; the plurality of first components are spliced on the second protective film to realize the splicing of the flexible device, thereby solving the problem of splicing difficulty in existing flexible devices.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to more clearly illustrate the technical solutions of the embodiments of the application, the drawings illustrating the embodiments will be briefly described below. Obviously, the drawings in the following description merely illustrate some embodiments of the present invention. Other

drawings may also be obtained by those skilled in the art according to these figures without paying creative work.

(2) In order to have a more complete understanding of the present application and its beneficial effects, the description will be given below in conjunction with the accompanying drawings. The same reference numerals in the following description indicate the same elements.

(3) FIG. 1 is a schematic flowchart of a method of manufacturing a spliced display screen provided by an embodiment of the present application.

(4) FIG. 2 is a schematic structural diagram corresponding to the manufacturing method shown in FIG. 1.

(5) FIG. 3 is a schematic structural diagram of a first component of the spliced display screen provided by an embodiment of the present application.

(6) FIG. 4 is a schematic structural diagram of the spliced display screen provided by an embodiment of the present application.

(7) FIG. 5 is a schematic diagram of arrangement of a plurality of second components on a second protective film.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(8) The technical solutions in the embodiments of the present application will be clearly and completely described in the following with reference to the accompanying drawings in the embodiments. It is apparent that the described embodiments are only a part of the embodiments of the present application, and not all of them. All other embodiments obtained by a person skilled in the art based on the embodiments of the present application without creative efforts are within the scope of the present application.

(9) In the description of this application, it should be understood that the terms “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “Rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, “clockwise”, “counterclockwise”, and the like are based on the orientation or positional relationship shown in the drawings, and is merely for the convenience of describing the present invention and simplifying the description, rather than indicating or implying that the device or element referred to must have a specific orientation, structure and operation in a specific orientation, which should not be construed as limitations on the present invention. In addition, the terms “first” and “second” are used for descriptive purposes only, and cannot be understood as indicating or implying relative importance or implicitly indicating the number of technical features indicated. Therefore, the features defined as “first” and “second” may explicitly or implicitly include one or more of the features. In the description of the present application, the meaning of “a plurality” is two or more, unless specifically defined otherwise.

(10) In manufacturing processes of at least part of flexible devices in flexible electronic products, a flexible substrate is often formed on a rigid glass substrate first, then related electronic devices are fabricated on the flexible substrate, and the glass substrate is separated from the flexible substrate after the fabrication of the related electronic devices is completed, thereby forming a flexible device with the flexible substrate as a base. To make a large-sized screen, a plurality of flexible devices need to be spliced together; however, it is difficult to splice flexible devices because their edges are prone to curling, and it is difficult to control splicing seams. In another approach, a flexible substrate is first formed on a rigid glass substrate, then related electronic devices are fabricated on the flexible substrate, then a display panels are formed after the fabrication of the related electronic devices is completed, and the plurality of display panels are spliced together followed by removing the glass substrate. However, there are technical problems in separating the glass substrate from the flexible substrate using laser lift-off technology after splicing.

(11) Therefore, in order to solve the above-mentioned problems, the present application proposes a method of manufacturing a spliced display screen and a display screen. The present application will be further described below in conjunction with the drawings and implementations.

(12) Referring to FIG. 1 and FIG. 2. FIG. 1 is a schematic flow diagram of a method of manufacturing a spliced display screen provided by an embodiment of the present application, and FIG. 2 is a schematic structural diagram corresponding to the method shown in FIG. 1. The embodiment of the present application provides a method of manufacturing a spliced display screen, and the specific process of the method of manufacturing a display module is as follows:

(13) **101.** providing a glass substrate and a display panel, wherein the display panel includes a flexible substrate and a display unit sequentially formed on a surface of a side of the glass substrate, and the display panel includes a bonding area and a non-bonding area.

(14) A glass substrate **20** is provided, and a display panel is prepared on the glass substrate **20**. The display panel includes a flexible substrate **10** and a display unit which are sequentially prepared on a surface of a side of the glass substrate **20**.

(15) In this embodiment, a material of the flexible substrate **10** is preferably a high-temperature resistant polyimide (PI) material. In other embodiments, the flexible substrate **10** may also be made of a material including silicon dioxide, polyester resin, polyethylene, polypropylene, polystyrene, polylactic acid, polyethylene terephthalate, or polyurethane.

(16) In this embodiment, the display unit is a mini-LED. In other embodiments, the display unit may further include: one of a light-emitting diode (LED), a micro-LED, or an organic light-emitting diode (OLED).

(17) The manufacturing of the display panel also includes: electrically connecting a chip-on-film (COF) **30** to one end of each of the flexible substrate **10**; and electrically connecting a printed circuit board (PCB) to an end of each of the chip-on-film away from the flexible substrate **40**.

(18) **102.** attaching a first protective film to a side of the display unit away from the glass substrate.

(19) A first protective film **50** is attached to the side of the display unit away from the glass substrate **20**, wherein a material of the first protective film **50** includes PET. By utilizing PET's good mechanical properties, oil resistance, fat resistance, dilute acid resistance, dilute alkali resistance, and resistance to most solvent, it can better protect the display panel underneath. In addition, by utilizing PET's high transparency, it can prevent impact on the light transmission performance of the display panel.

(20) A cutting mark is disposed on the first protective film **50**, and it is appreciated that the cutting mark is provided in a non-bonding area.

(21) **103.** disposing a supporting glass on a side of the first protective film away from the flexible substrate.

(22) The supporting glass **70** is connected to the first protective film **50** by a low-viscosity glue **60** at the side of the first protective film **50** away from the flexible substrate **10**. The supporting glass **70** is mainly configured to support the flexible substrate **10** and the display unit after the glass substrate **20** is separated from the flexible substrate **10**, so as to prevent a phenomenon of warping of the flexible substrate **10**.

(23) **104.** separating the glass substrate from the flexible substrate.

(24) The glass substrate **20** is separated from the flexible substrate **10** by means of laser peeling. Exemplarily, a laser beam is adopted to irradiate the display screen to be peeled off, and the laser beam moves relative to the display screen to be peeled off, thereby separating the glass substrate **20** from the flexible substrate **10** on the display screen to be peeled off.

(25) **105.** cutting the first protective film, the display panel, and the supporting glass located in the non-bonding area to obtain a plurality of first components.

(26) Using laser cutting, the first protective film **50**, the display panel, and the supporting glass **70** located in the non-bonding area are cut according to the cutting mark reserved on the first protective film **50**; and the first protective film **50**, the display panel, and the supporting glass **70** that have been cut are defined as the first component **100**. A specific structure of the first component **100** can be referred to FIG. 3, which is a schematic diagram of a first component **100** of the spliced display screen provided by an embodiment of the present application.

(27) The end of the cutfirst protective film **50** away from the bonding area, the end of the flexible substrate **10** away from the bonding area, and the end of the supporting glass **70** away from the bonding area are flush with each other.

(28) **106.** providing a second protective film, and splicing the plurality of first components **100** on a surface of a side of the second protective film, wherein the second protective film is disposed on a side of the flexible substrate away from the first protective film.

(29) The second protective film is provided, wherein a material of the second protective film includes PET. By utilizing PET's good mechanical properties, oil resistance, fat resistance, dilute acid resistance, dilute alkali resistance, and resistance to most solvent, it can better protect the display panel underneath. In addition, by utilizing PET's high transparency, it can prevent impact on the light transmission performance of the display panel.

(30) A plurality of first components **100** are spliced on the surface of one side of the second protective film **80**. Exemplarily, an end of one of the first components **100** away from the bonding area is arranged adjacent to an end of another one of the first components **100** away from the bonding area. It should be noted that there are splicing seams between adjacent first components **100**, but since sizes of the splicing seams and distances between the pixels are not much different, visually seamless splicing can be achieved, thereby improving the quality of flexible splicing products.

(31) It is appreciated that in order to realize that there is no chip-on-films **30** and printed circuit boards **40** at the splicing seams, the chip-on-films **30** of the display panels of two of the first components **100** in each row are all located at one end of the display units electrically connected to the chip-on-films **30** away from the splicing seams.

(32) In some embodiments, after the plurality of first components **100** are spliced on the surface of one side of the second protective film **80**, the supporting glass **70** is separated from the first protective film **50** to obtain a spliced flexible mini-LED screen, which can be visually seamless.

(33) In the method of manufacturing the spliced display screen provided by the embodiment of the present application, the supporting glass **70** is disposed on the first protective film **50**, so that the flexible substrate **10** can be separated from the glass substrate **20** before the plurality of flexible substrates **10** are spliced on the second protective film **80**, and the supporting glass **70** is configured to support the flexible substrate **10** and the display unit, thereby preventing a phenomenon of warping after the flexible substrate **10** is separated from the glass substrate **20**. After the plurality of flexible substrates **10** are spliced on the second protective film **80**, the flexible substrate can be separated from the glass substrate **20**, thereby realizing seamless splicing of the plurality of flexible substrates **10**, which solves the problem of splicing difficulty in existing flexible devices.

(34) Still referring to FIG. 4, FIG. 4 is a schematic structural diagram of a spliced display screen provided by an embodiment of the present application. Still another embodiment of the present application also provides a spliced display screen **1**. The spliced display screen **1** includes a second protective film **80** and a plurality of first components **100** arranged adjacently on the second protective film **80**, wherein the plurality of first components **100** are arranged on a same side of the second protective film **80** to form a plurality of second components **200**, each of the first components **100** includes a display panel and a first protective film **5** that are stacked, the display panel is disposed between the first protective film **50** and the second protective film **80**, wherein the display panel includes a flexible substrate **10** and a display unit that are stacked.

(35) An end of the first protective film **50** away from a bonding area is flush with an end of the flexible substrate **10** away from the bonding area. By arranging an edge of the first protective film **50** to be flush with an edge of the flexible substrate **10**, the splicing seams between adjacent ones of the first components **100** can be made smaller, thereby making it possible to achieve visually seamless.

(36) The first protective film **50** and the second protective film **80** are both PET films, which can better protect the display panel underneath.

(37) A chip-on-film **30** is disposed at one end of each of the flexible substrate **10** away from the non-bonding area; and a printed circuit board **40** is disposed at one end of each of the chip-on-film **30** away from the flexible substrate **10**. In order to realize that there is no chip-on-films **30** and printed circuit boards **40** at the splicing seams, the chip-on-films **30** of the display panels of two of the first components **100** in each row are all located at one end of the display units electrically connected to the chip-on-films away from the splicing seams. For example, in this embodiment, the chip-on-film of the first component **100** on the left is located at a left end of the display unit electrically connected to the chip-on-film **30**, and the chip-on-film **30** of the first component **100** on the right is located at a right end of the display unit electrically connected to the chip-on-film **30**, thereby ensuring that there is no chip-on-film **30** at a splicing seam. The printed circuit board **40** of the first component **100** on the left is located at the left end of the display unit electrically connected to the printed circuit board **40**, and the printed circuit board **40** of the first component **100** on the right is located at the right end of the display unit electrically connected to the printed circuit board **40**, thereby ensuring that there is no printed circuit board **40** at the splicing seam.

(38) The display unit is a mini-LED. In other embodiments, the display unit may further include: one of a light-emitting diode (LED), a micro-LED, or an organic light-emitting diode (OLED).

(39) Still referring to FIG. 5, FIG. 5 is a schematic diagram of arrangement of a plurality of second components on a second protective film. A number of rows of the plurality of second components **200** spliced on the second protective film **80** is n , a number of columns of the plurality of second components **200** spliced on the second protective film **80** is 2 , and $n \geq 1$. In this embodiment, $n=6$. In other embodiments, the number of rows of the plurality of second components **200** spliced on the second protective film **80** may be 2 , the number of columns of the plurality of second components **200** spliced on the second protective film **80** may be n , and $n \geq 1$. By splicing the plurality of second components on the second protective film, flexible electronic products with large-sized screens can be realized by splicing the plurality of small flexible display panels, which can not only retain advantages of flexible display screens, but also realize large-sized screens. In addition, in the embodiment of the present application, the plurality of second components are seamlessly spliced on the second protective film, which can improve the display quality and bring a better experience to customers.

(40) The method of manufacturing the spliced display screen and the display screen provided by the embodiments of the present application are described in detail above. Specific examples are used to explain the principle and implementation of the present application. The descriptions of the above embodiments are only used to help understand the present application. Also, for those skilled in the art, according to the ideas of the present application, there will be changes in the specific implementation and application scope. In summary, the content of this specification should not be construed as limiting the present application.

Claims

1. A method of manufacturing a spliced display screen, wherein the manufacturing method comprises: providing a glass substrate and a display panel, wherein the display panel comprises a flexible substrate and a display unit sequentially formed on a surface of a side of the glass substrate, and the display panel comprises a bonding area and a non-bonding area; attaching a first protective film to a side of the display unit away from the glass substrate; disposing a supporting glass on a side of the first protective film away from the flexible substrate; separating the glass substrate from the flexible substrate; cutting the first protective film, the display panel, and the supporting glass located in the non-bonding area to obtain a plurality of first components; providing a second protective film, and splicing the plurality of first components on a surface of a side of the second protective film, wherein the second protective film is disposed on a side of the flexible substrate away from the first protective film.

2. The manufacturing method according to claim 1, wherein the step of splicing the plurality of first components on the surface of the side of the second protective film comprises: disposing an end of one of the first components away from the bonding area adjacent to an end of another one of the first components away from the bonding area.
 3. The manufacturing method according to claim 1, wherein after the step of splicing the plurality of first components on the surface of the side of the second protective film, the manufacturing method further comprises: separating the supporting glass from the first protective film.
 4. The manufacturing method according to claim 1, wherein the manufacturing method further comprises: electrically connecting a chip-on-film to one end of each of the flexible substrate located in the bonding area; and electrically connecting a printed circuit board to an end of each of the chip-on-film away from the flexible substrate.
 5. The manufacturing method according to claim 4, wherein splicing seams are defined between adjacent ones of the first components.
 6. The manufacturing method according to claim 5, wherein each of chip-on-films of display panels of two of the first components in each row is all disposed at one end of the display unit electrically connected to the chip-on-films away from the splicing seams.
 7. The manufacturing method according to claim 1, wherein an end of the first protective film away from the bonding area is flush with an end of the flexible substrate away from the bonding area.
 8. The manufacturing method according to claim 1, wherein a number of rows of the plurality of second components spliced on the second protective film is n ; a number of columns of the plurality of second components spliced on the second protective film is 1; and the $n > 1$.
 9. The manufacturing method according to claim 1, wherein a number of rows of the plurality of second components spliced on the second protective film is 2; a number of columns of the plurality of second components spliced on the second protective film is n ; and the $n > 1$.
 10. The manufacturing method according to claim 1, wherein the first protective film and the second protective film are both polyethylene terephthalate (PET) films.
 11. The manufacturing method according to claim 1, wherein the display unit comprises one of LED, mini-LED, micro-LED, or OLED.
 12. The manufacturing method according to claim 1, wherein the supporting glass is connected to the first protective film by a low-viscosity glue at the side of the first protective film away from the flexible substrate.
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